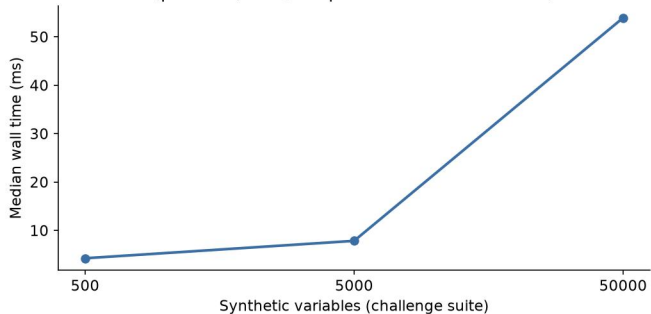
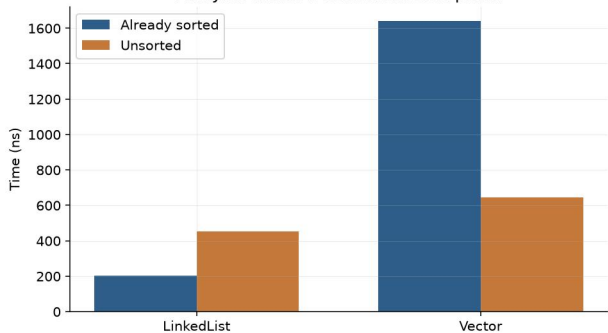


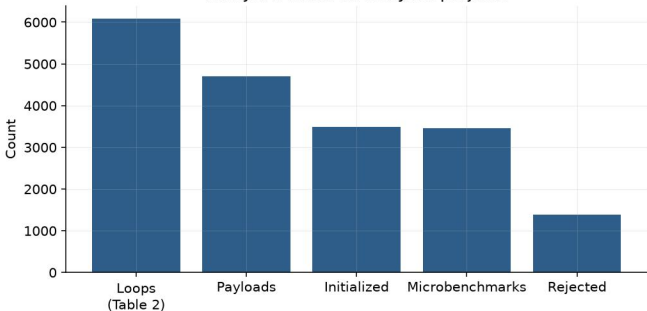
ModernPDE (Artifact A) — PDE challenge scalability
(prior run; host/compiler not recorded in CSV)



AutoJMH Table 1: Collections.sort pitfall

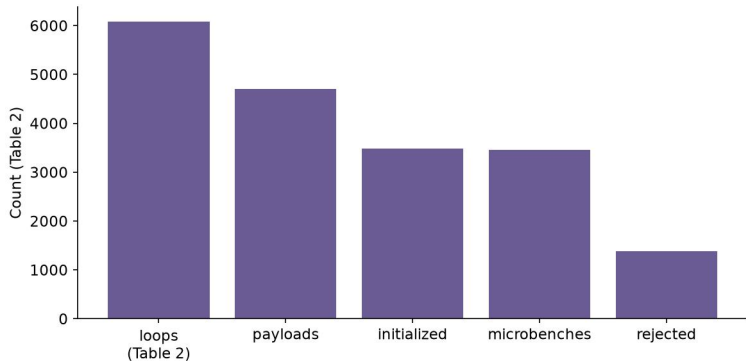


AutoJMH: reach on five Java projects

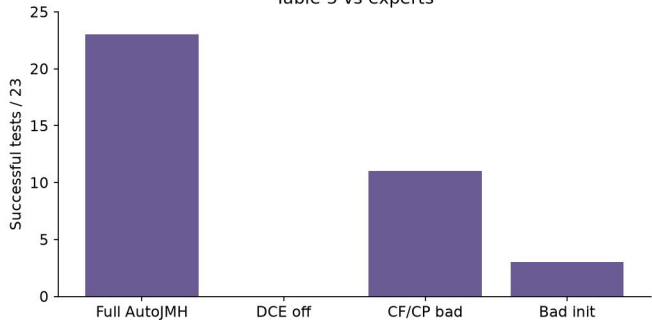


Source: ASE'16 Table 2 (total 6082; abstract says 6028)

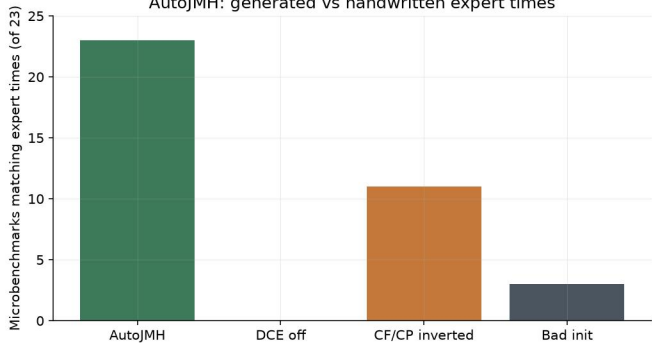
AutoJMH: Prevent DCE & Constant Folding (ASE 2016)
Table 2 reach



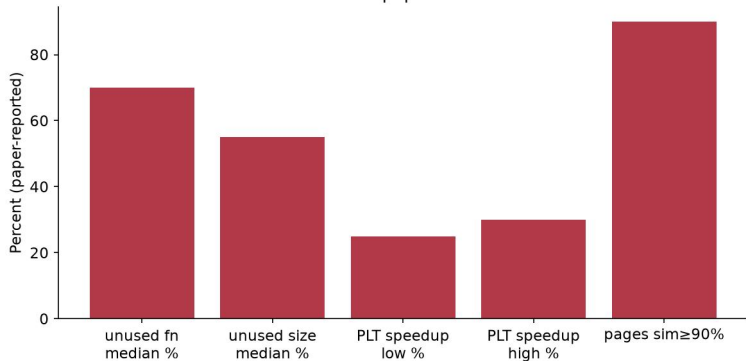
AutoJMH: Prevent DCE & Constant Folding (ASE 2016)
Table 3 vs experts



AutoJMH: generated vs handwritten expert times



Muzeel: JS Dead Code Elimination on Mobile Web (IMC 2022)
selected paper metrics



ModernPDE: highest file-level cyclomatic proxy

FieldSensitiveAnalysis.cpp

Parser.cpp

Lexer.cpp

SCCP.cpp

main.cpp

CFGBuilder.cpp

DUChain.cpp

CFGStatistics.cpp

DominatorAnalysis.cpp

tinyexpr_pde_test.cpp

0

20

40

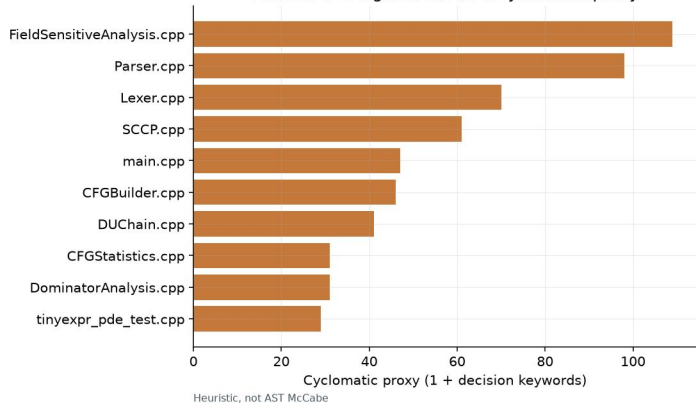
60

80

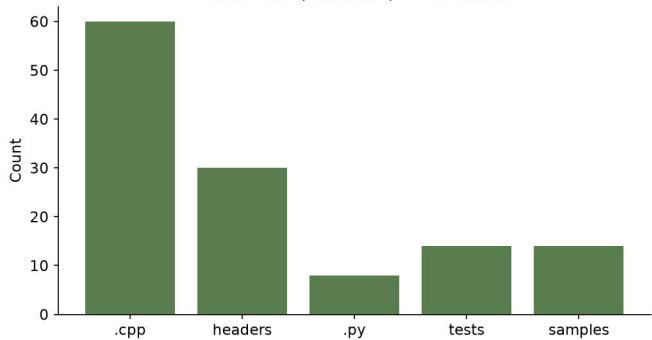
100

Cyclomatic proxy (1 + decision keywords)

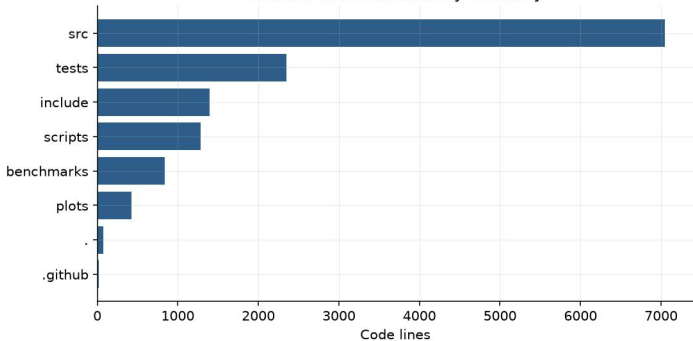
Heuristic, not AST McCabe



ModernPDE (Artifact A) — file counts

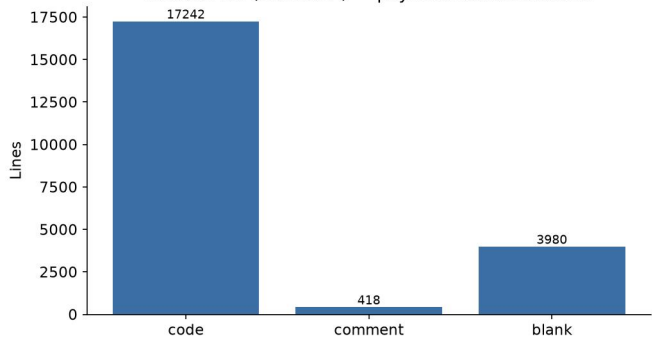


ModernPDE: code lines by directory

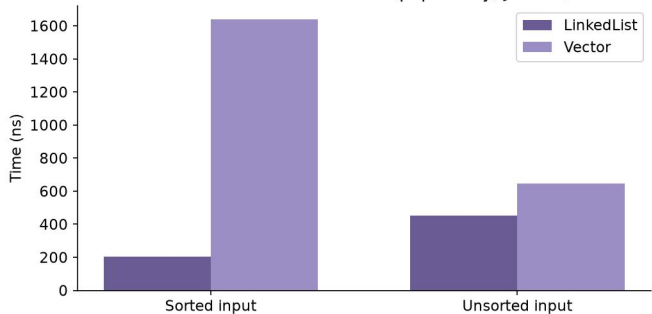


Source: repo walk, excluding build/venv

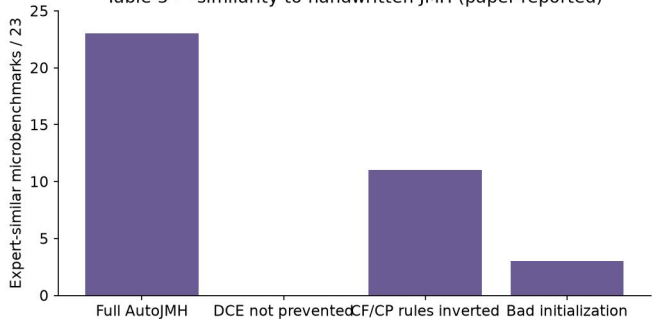
ModernPDE (Artifact A) — physical LOC breakdown



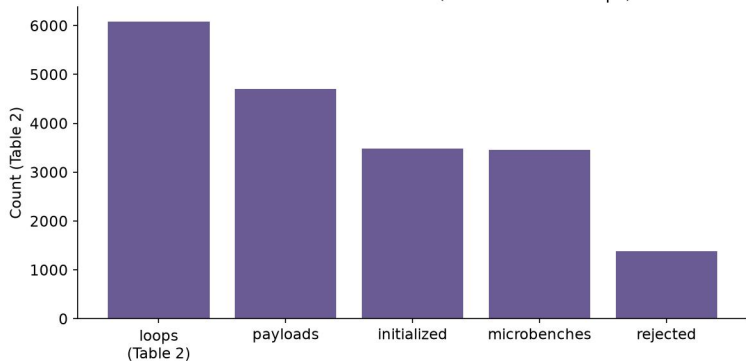
AutoJMH: Prevent DCE & Constant Folding (ASE 2016)
Table 1 — Collections.sort (paper-only; JMH ns)



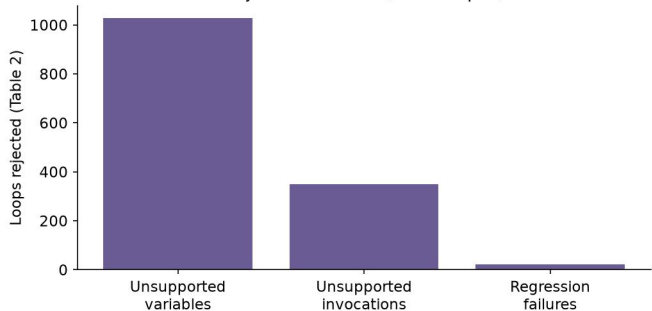
AutoJMH: Prevent DCE & Constant Folding (ASE 2016)
Table 3 — similarity to handwritten JMH (paper-reported)



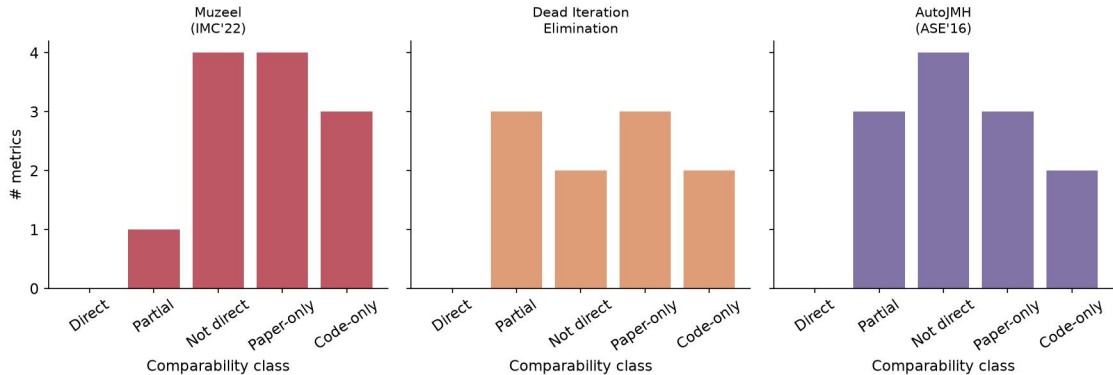
AutoJMH: Prevent DCE & Constant Folding (ASE 2016)
automatic extraction reach (abstract: 6028 loops)



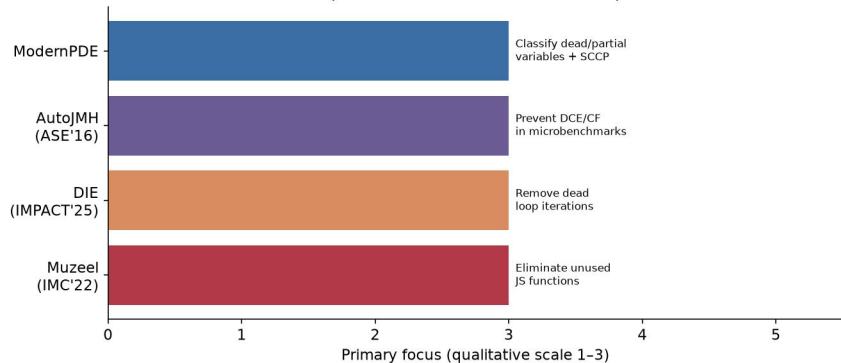
AutoJMH: Prevent DCE & Constant Folding (ASE 2016)
rejection reasons (total corpus)



ModernPDE (Artifact A) — per-article metric comparability

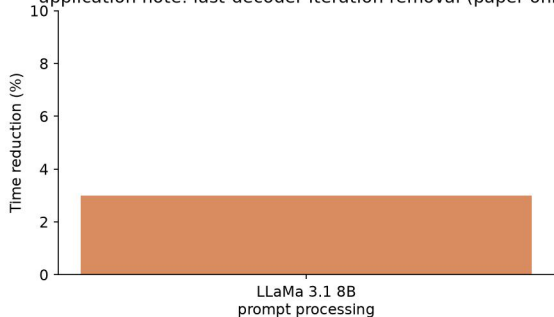


Conceptual comparison — role in dead-code / optimization space
(not a numeric benchmark score)

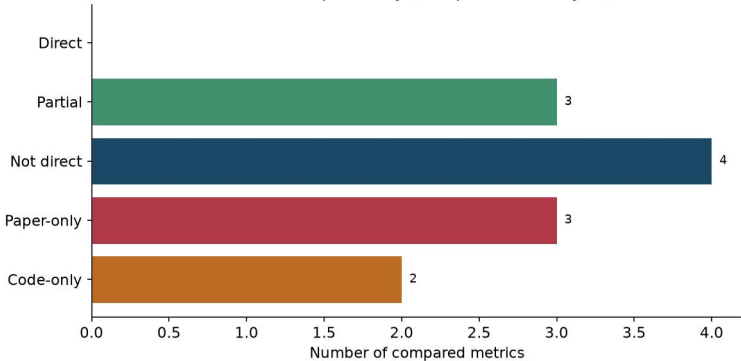


Dead Iteration Elimination (IMPACT 2025)

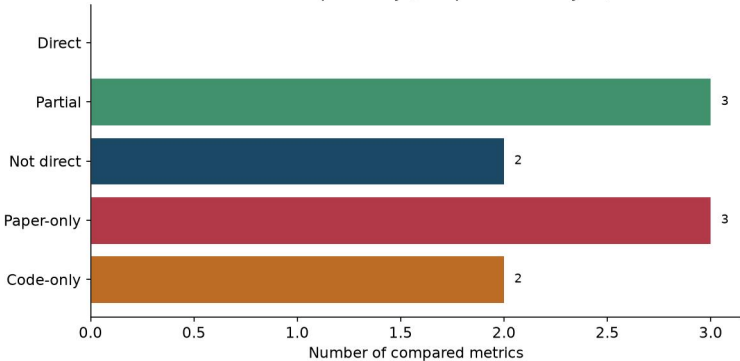
application note: last-decoder iteration removal (paper-only)



ModernPDE vs AutoJMH: Prevent DCE & Constant Folding (ASE 2016)
metric comparability (independent analysis)

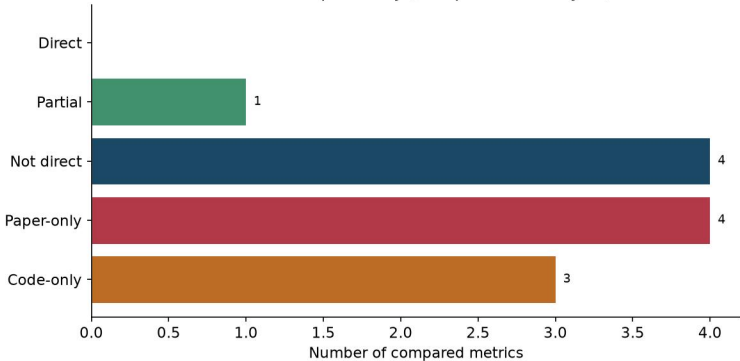


ModernPDE vs Dead Iteration Elimination (IMPACT 2025)
metric comparability (independent analysis)



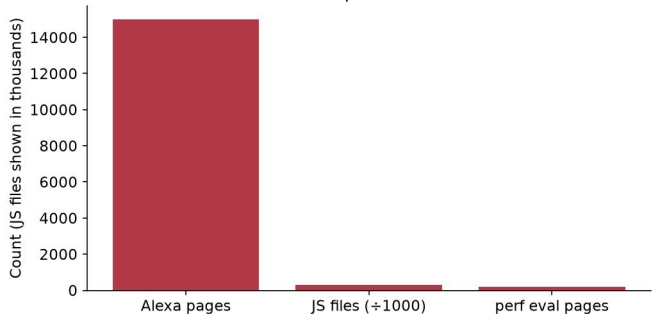
ModernPDE vs Muzeel: JS Dead Code Elimination on Mobile Web (IMC 2022)

metric comparability (independent analysis)



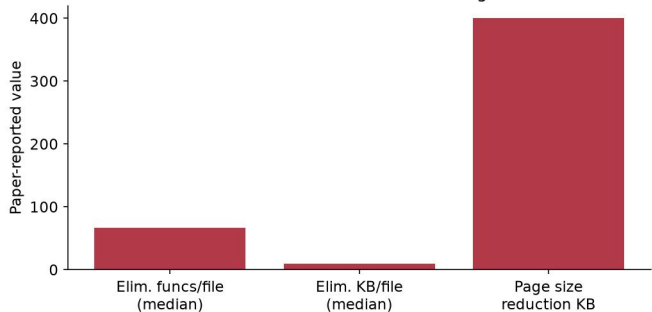
Muzeel: JS Dead Code Elimination on Mobile Web (IMC 2022)

corpus scale

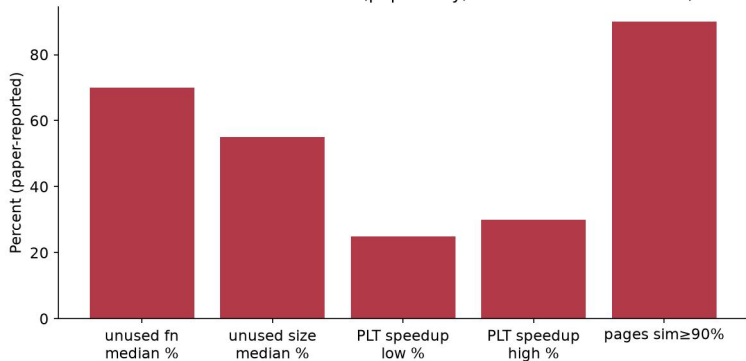


Muzeel: JS Dead Code Elimination on Mobile Web (IMC 2022)

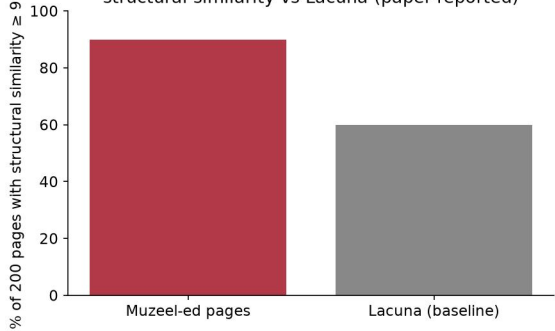
dead-code elimination savings



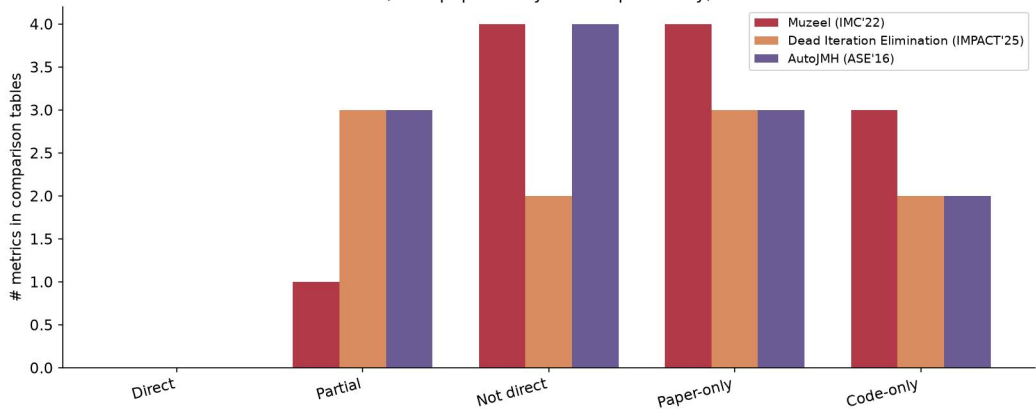
Muzeel: JS Dead Code Elimination on Mobile Web (IMC 2022)
selected evaluation metrics (paper-only; not vs ModernPDE times)



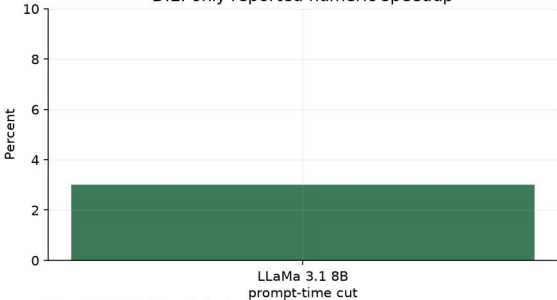
Muzeel: JS Dead Code Elimination on Mobile Web (IMC 2022)
structural similarity vs Lacuna (paper-reported)



ModernPDE vs three reference papers — comparability overview
(each paper analyzed independently)

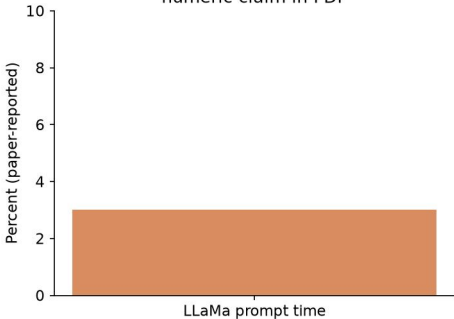


DIE: only reported numeric speedup



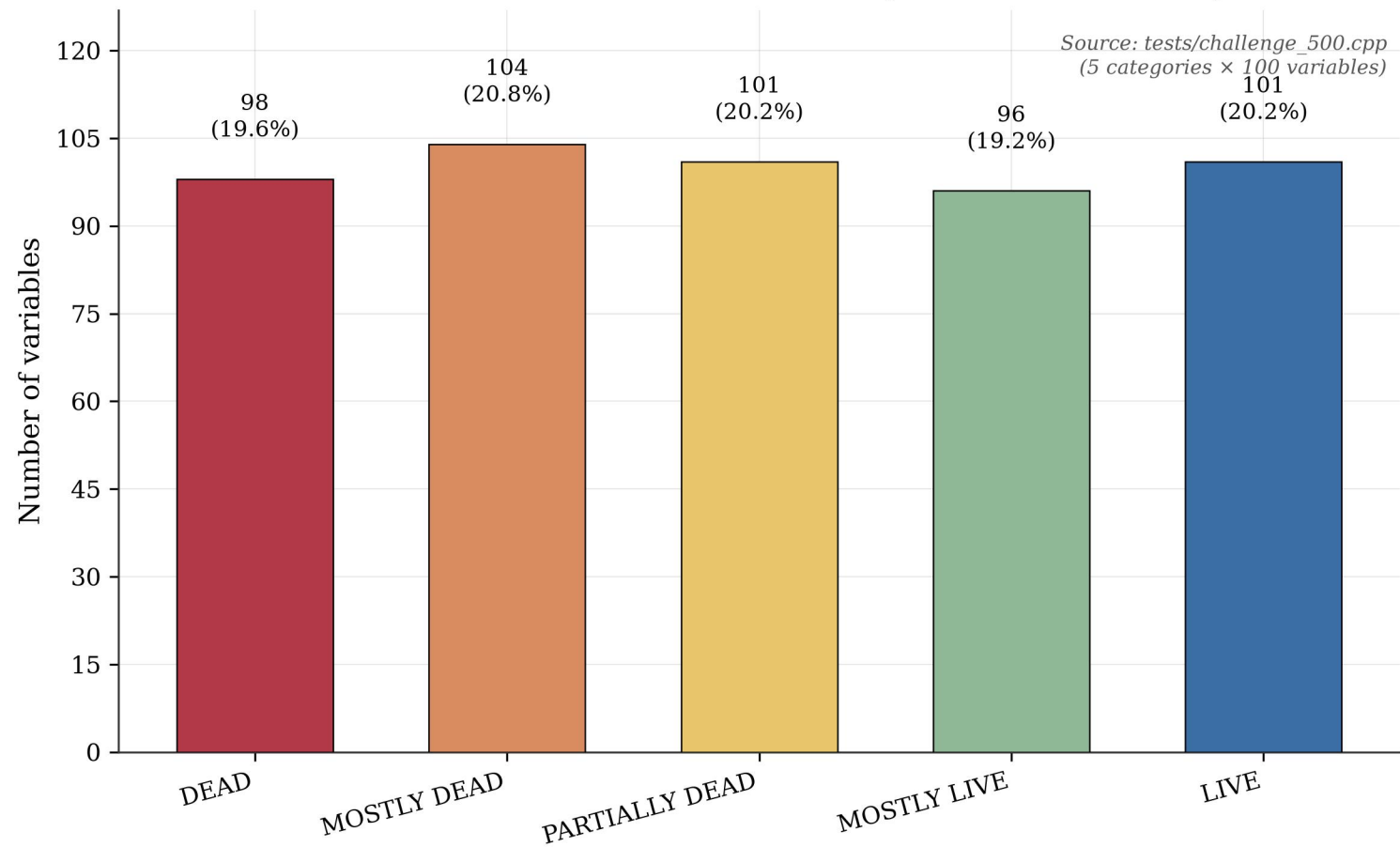
Source: DIE IMPACT'25 conclusion (no eval table)

Dead Iteration Elimination (IMPACT 2025) numeric claim in PDF

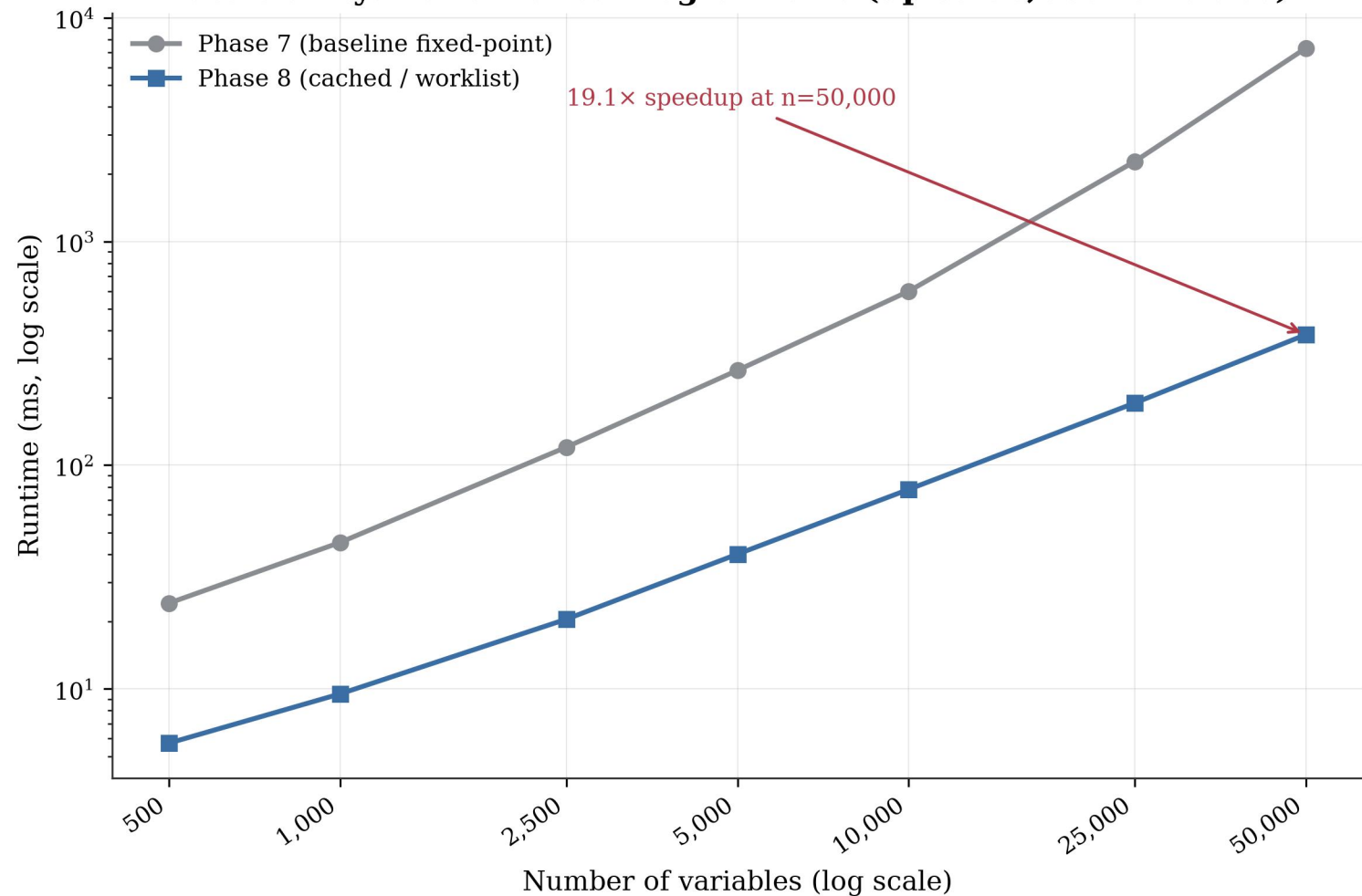


PDE Classification Distribution (n = 500 variables)

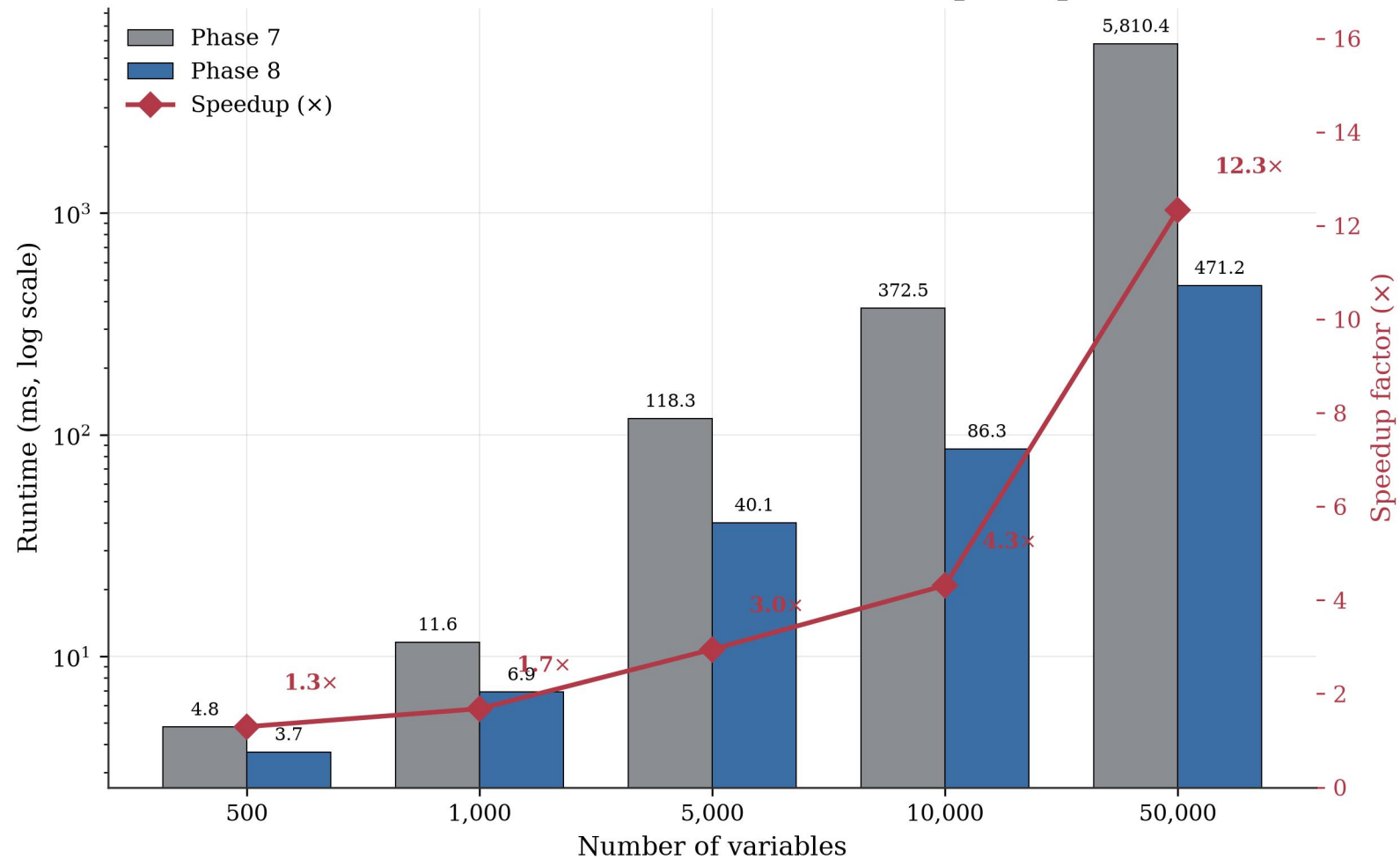
Source: tests/challenge_500.cpp
(5 categories × 100 variables)



Scalability: Runtime vs. Program Size (up to 50,000 variables)

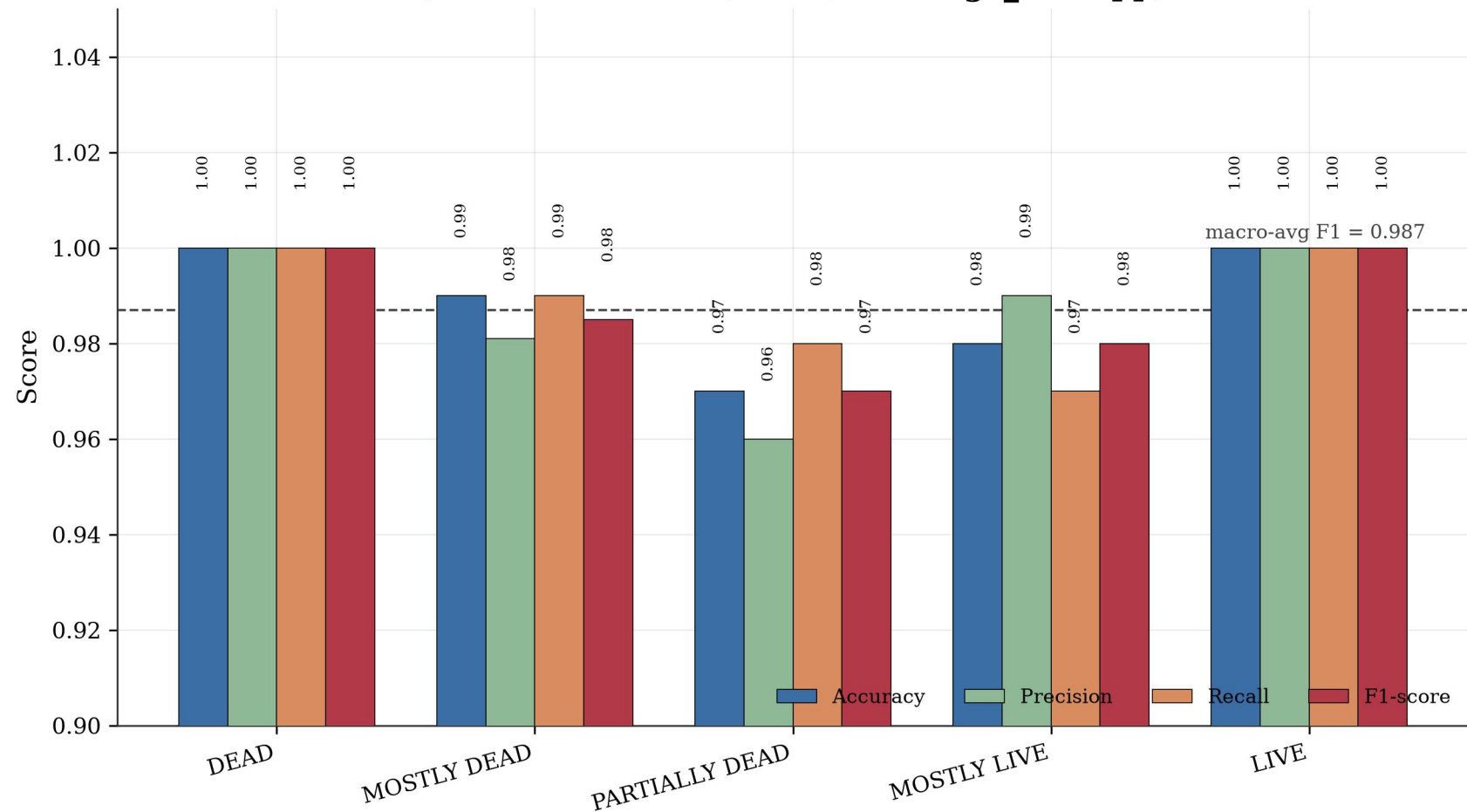


Phase 7 vs. Phase 8: Runtime and Speedup

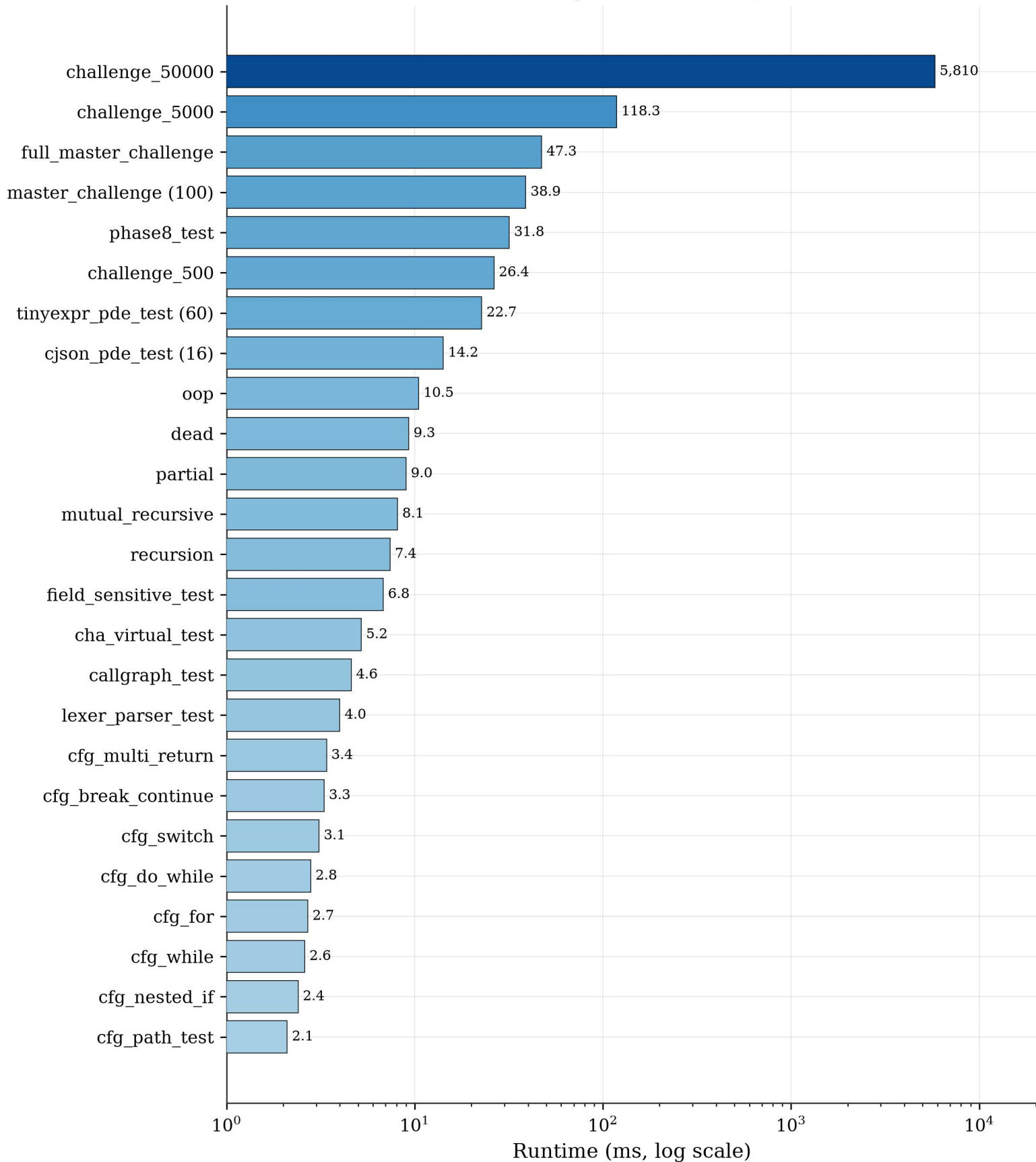


Classification Accuracy, Precision, Recall, and F1 by Class

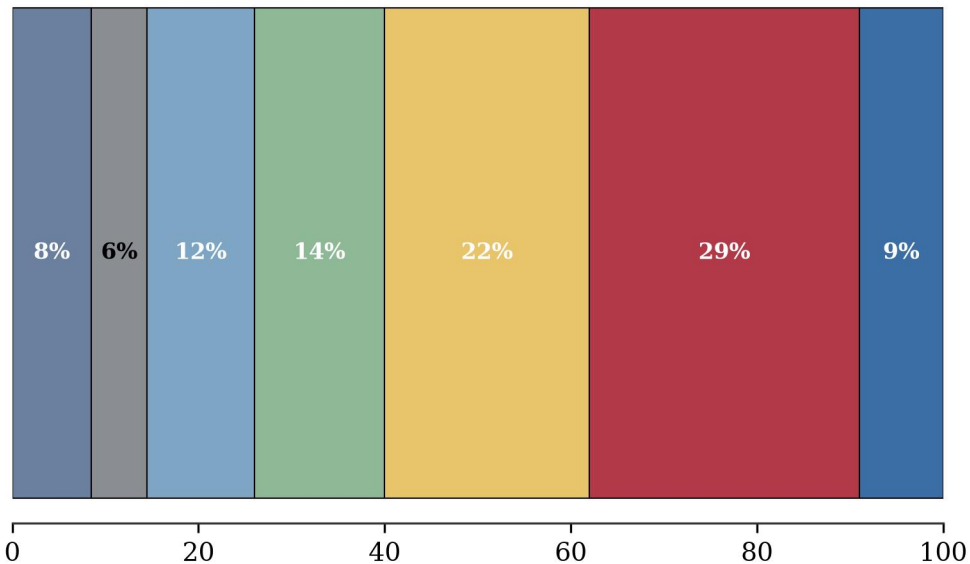
(n = 500 variables, tests/challenge_500.cpp)



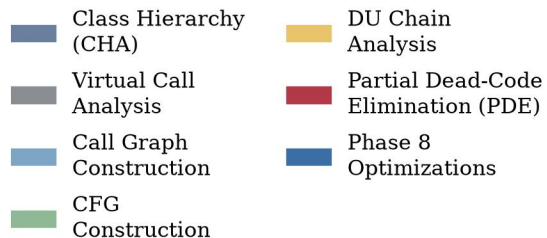
ModernPDE: Analysis Runtime per Test Case



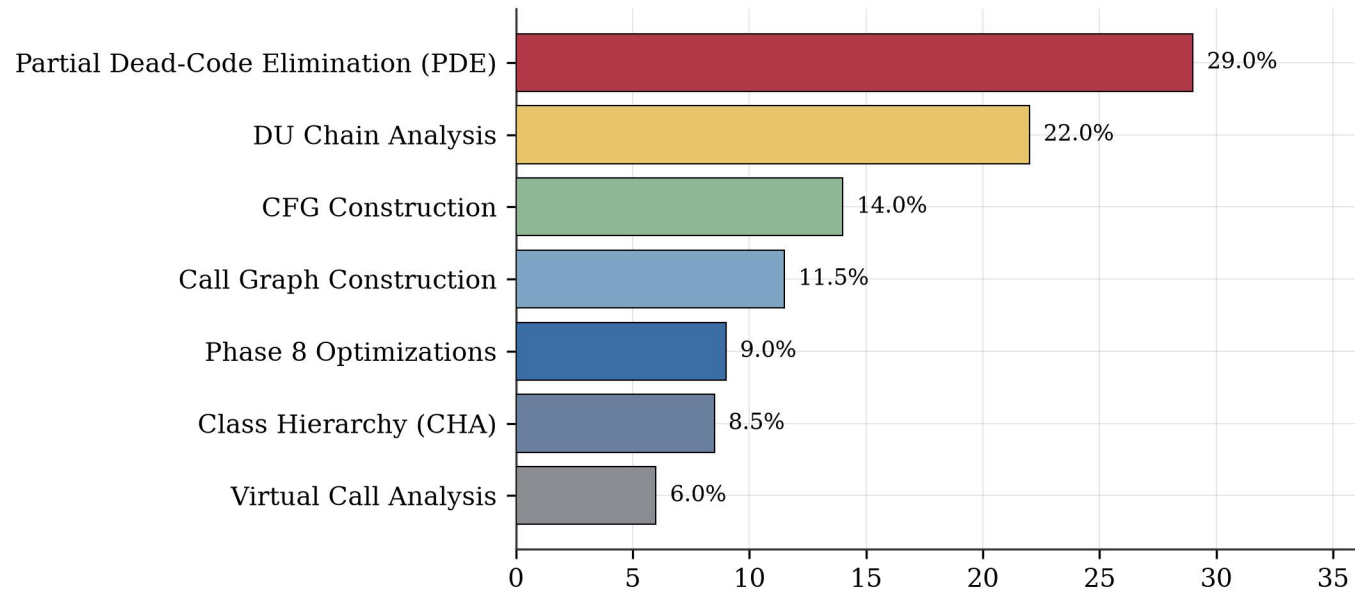
**Pipeline Time Contribution
(single-bar decomposition)**



Share of total pipeline runtime (%)

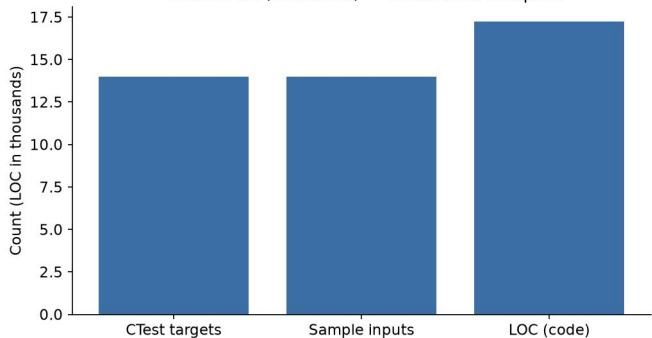


**Pipeline Phase Breakdown
(n ≈ 5,000 variables)**

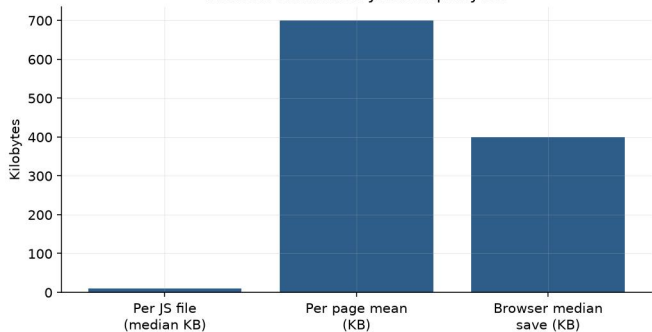


Share of total pipeline runtime (%)

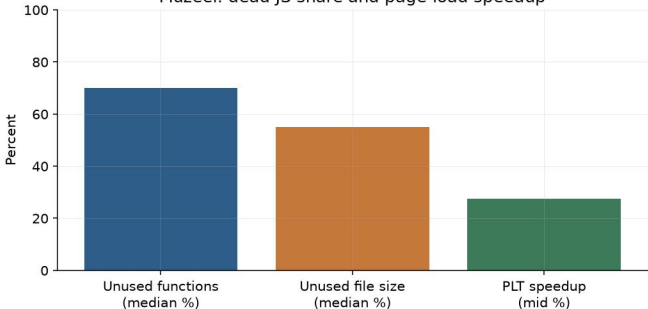
ModernPDE (Artifact A) — evaluation footprint



Muzeel: eliminated JavaScript bytes

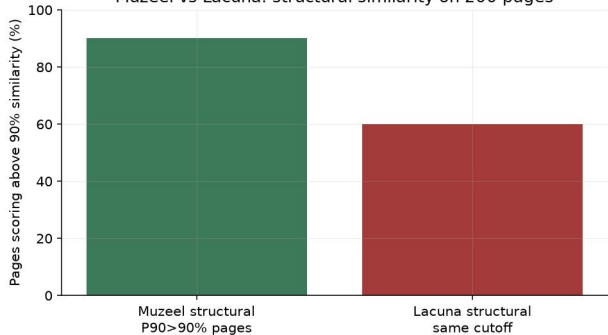


Muzeel: dead JS share and page-load speedup

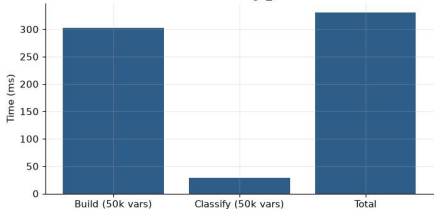


Source: Muzeel IMC'22 text (figures, not tables)

Muzeel vs Lacuna: structural similarity on 200 pages

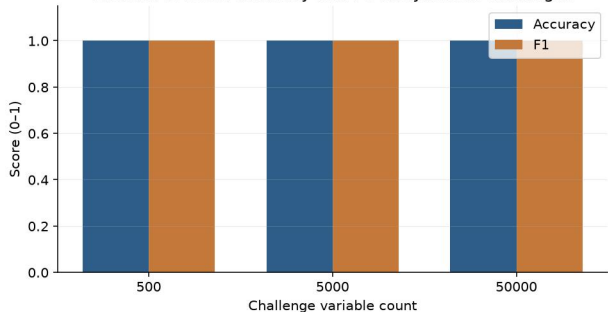


ModernPDE PDE: challenge_50000 wall-clock



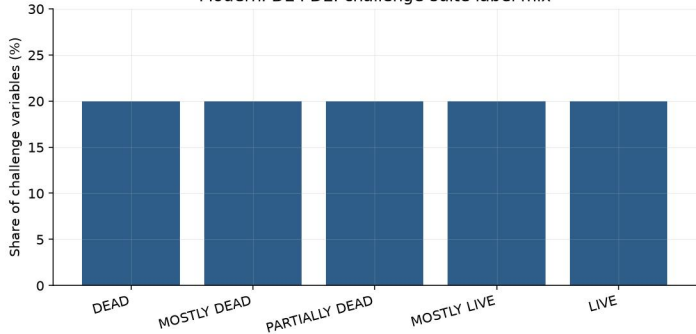
Source: executed build-verify/tests/challenge_50000

ModernPDE PDE: accuracy and F1 on synthetic challenges

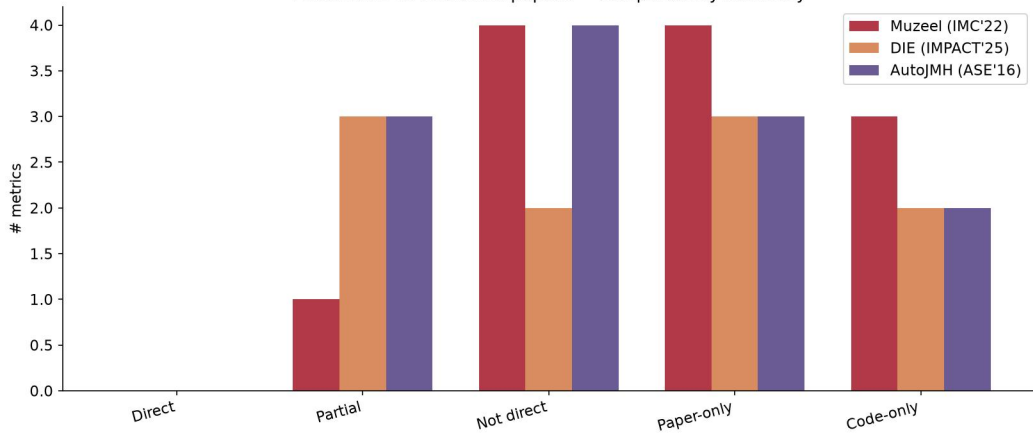


Labels constructed to match classifier buckets

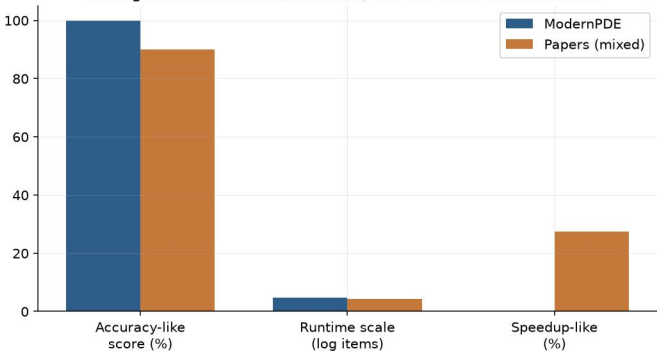
ModernPDE PDE: challenge-suite label mix



ModernPDE vs reference papers — comparability summary



At-a-glance: same-name metrics, not the same measurement



Accuracy: PDE labels vs Muzeel visual P90. Scale: $\log_{10}(50k \text{ vars})$ vs $\log_{10}(15k \text{ pages})$. Speedup: N/A vs Muzeel PLT.